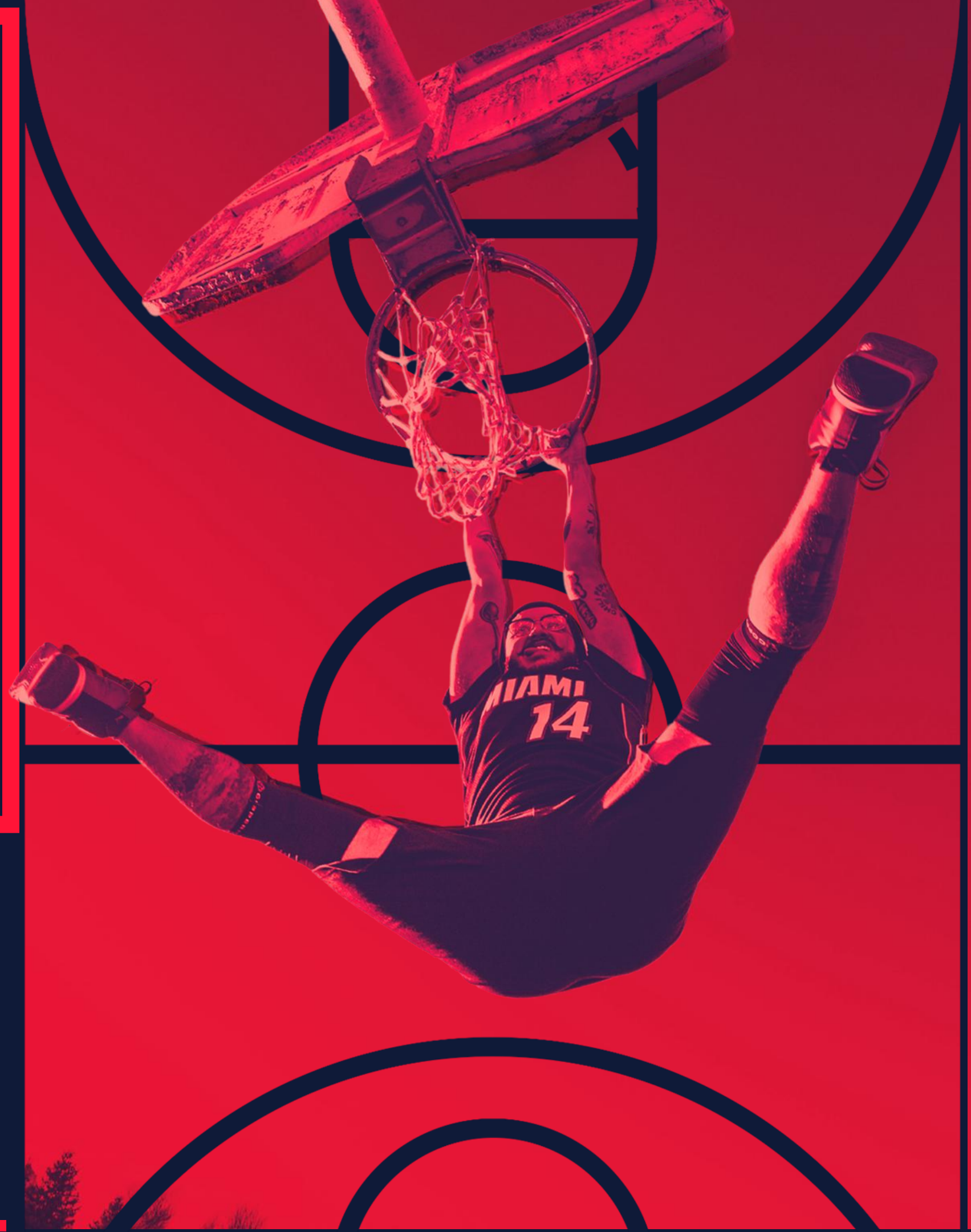


BASKET BALL

**Comprehensive Analysis
of NBA League Data**



Members

Mentor : Mr.MohammadhosseinTajAdini

Mr.SadeghMahmoudAbadi

Ms.DonyaEsfandiyarPour

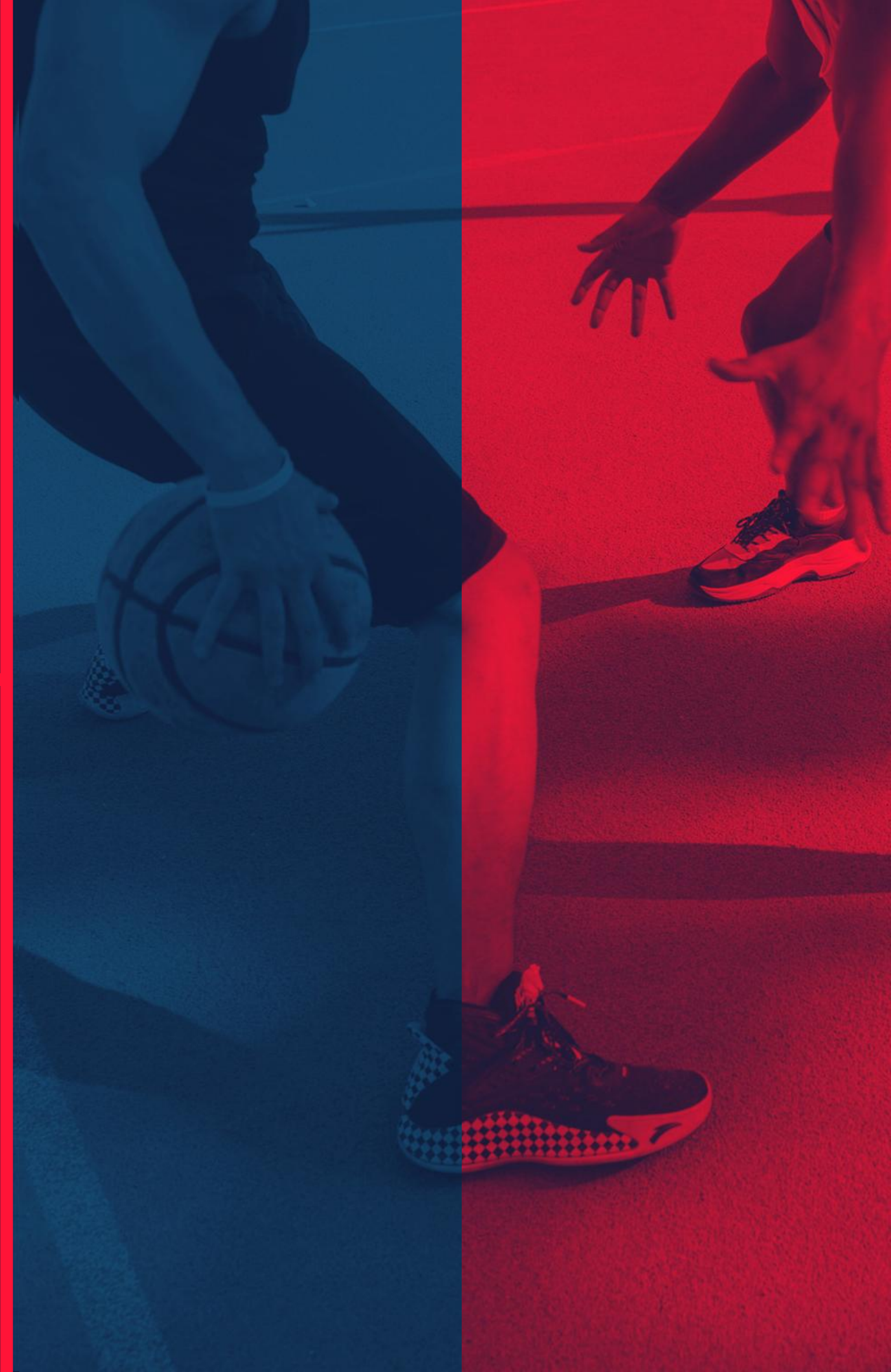
Ms.SagharFarahani

Mr.RozbehAbazari

Ms.MoonesLatifi

AGENDA

- 01** Data Extraction
- 02** Relational Modeling
- 03** KPI Engineering & Descriptive Analysis
- 04** Hypothesis Testing



content

From Data to Front-Office Decisions

"In modern professional sports—especially the NBA—championships aren't won solely on the hardwood; they are built through data-driven strategies in the front office. Raw data alone is meaningless until we transform it into actionable KPIs and domain-specific metrics. In this project, we built a complete data pipeline to extract, structure, and model five seasons of NBA data. Our core objective is to deliver the direct answers to our key research questions: from analyzing physical distributions and testing core agility hypotheses, to delivering our final, data-backed recruitment decisions for the club. Let's dive straight into our methodology, engineered metrics, and final analytical results..

Data Extraction

Extracted player demographics, club metrics, and trophy histories (2019–2024).

Relational Modeling

Normalized.

KPI Engineering & Descriptive Analysis

Formulated domain metrics to evaluate player profiles and guide recruitment.

Hypothesis Testing

Statistical validation of player agility and experience efficiency.



IT'S GAME TIME!

1st Quarter : Data Extraction



Time Horizon / Scope

5 Consecutive Seasons (2019/20 – 2023/24)

- Player-Level Variables
- Coaches-Level Variables

ALL
NCE

Enter Person, Team, Section, etc

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2025-26 NBA Standings

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East	W	L	West	W	L
DET (1)	E	\$ 60 22	OKC (1)	E	\$ 64 18
BOS (2)	E	\$ 56 26	SAS (2)	E	\$ 62 20
NYK (3)	E	\$ 53 29	DEN (3)	E	\$ 54 28
CLE (4)	E	\$ 52 30	LAL (4)	E	\$ 53 29
TOR (5)	E	\$ 46 36	HOU (5)	E	\$ 52 30
ATL (6)	E	\$ 46 36	MIN (6)	E	\$ 49 33
PHI (7)	E	\$ 45 37	PHO (7)	E	\$ 45 37
ORL (8)	E	\$ 45 37	POR (8)	E	\$ 42 40
CHO (9)	E	\$ 44 38	LAC (9)	E	\$ 42 40
MIA (10)	E	\$ 43 39	GSW (10)	E	\$ 37 45
MIL (11)	E	\$ 32 50	NOP (11)	E	\$ 26 56
CHI (12)	E	\$ 31 51	DAL (12)	E	\$ 26 56
BRK (13)	E	\$ 20 62	MEM (13)	E	\$ 25 57
IND (14)	E	\$ 19 63	SAC (14)	E	\$ 22 60
WAS (15)	E	\$ 17 65	UTA (15)	E	\$ 22 60

Select Team Page:
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Identifiers & Context

Player_xp
team_id
player_id
Season_id



Scoring & Efficiency Metrics

Pts
Fg_pct
Efg_pct
Three_p_pct



Playmaking & Defense Features

Ast
Stl
Blk
tov

GOALS

Our main goal with these player variables was to look beyond just points scored. We focused on three core areas:

First, scoring efficiency—to see how well they shoot.

Second, total game impact—including defense, rebounds, and assists.

And third, win contribution—to measure how a player's stats directly lead to team wins



coaches-Level Variables

2024 EVENT GALLERY



- **Dimension Table**

Coach_id
name
Birthdate
nationality

- **Face Table**

Coach_id
Season_id
Team_id
Wins
losses





IT'S GAME TIME!

2nd Quarter : Relational Modeling

BASKETBALL



Entities

- teams
- Players
- Coaches
- Seasons
- awards

Relationships and Multivalued Attributes

- player_position
- award_season

Performance & Analytics

- player_stats
- team_stats
- coach_stats



IT'S GAME TIME!

**3rd Quarter : KPI Engineering &
Descriptive Analysis**

KPI

Measurable goals for success

Player

- Player efficiency based on age
- Assists vs Field goals
- dependancy on players (variety of points for players)
- xp / age (the more the better)
- offensive vs defensive rebounds

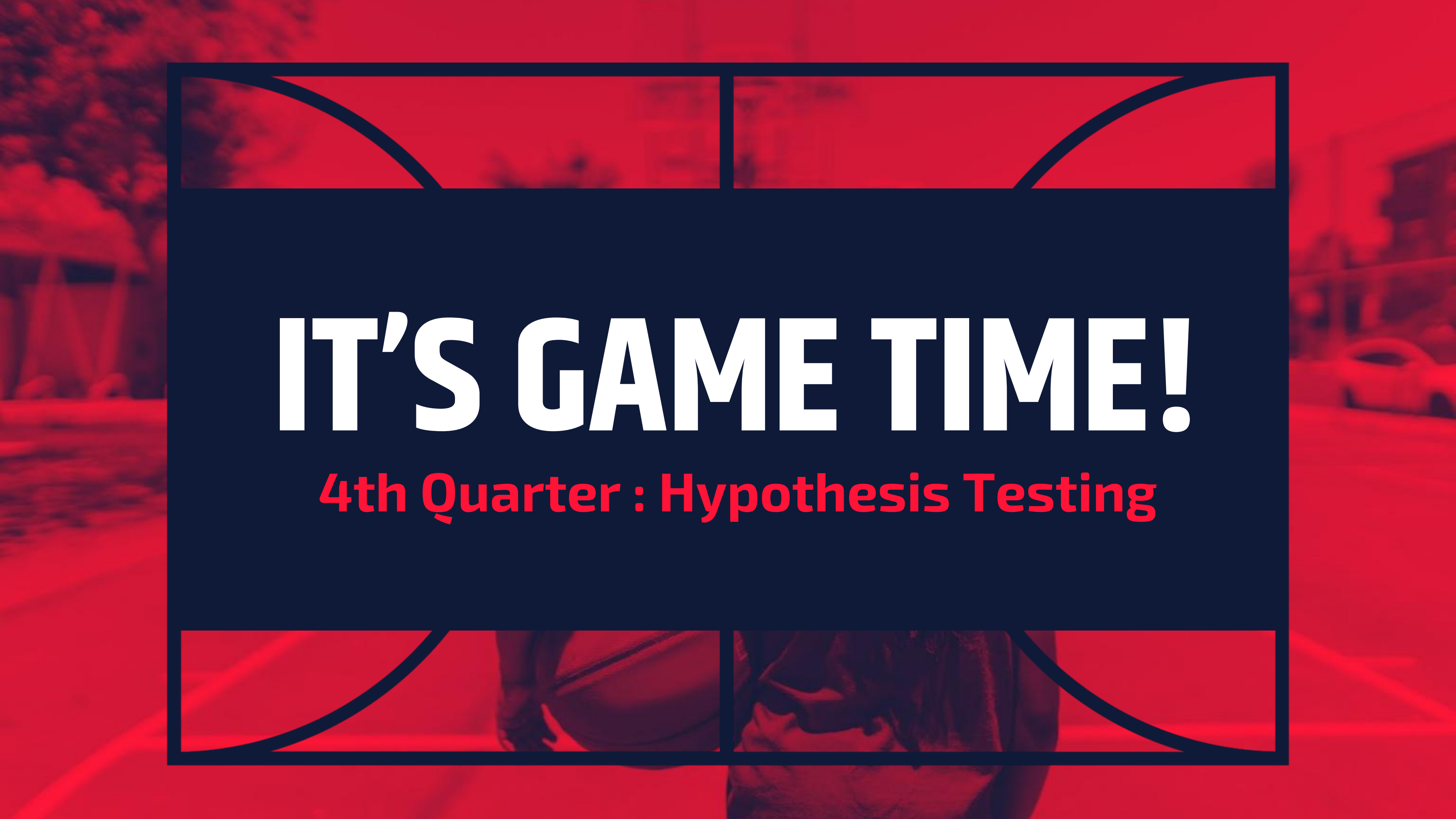
Coach

- wins / losses per season
- offensive_focus
- championship_probability
- player performance under coach(win / per)
- offensive_focus

Teams

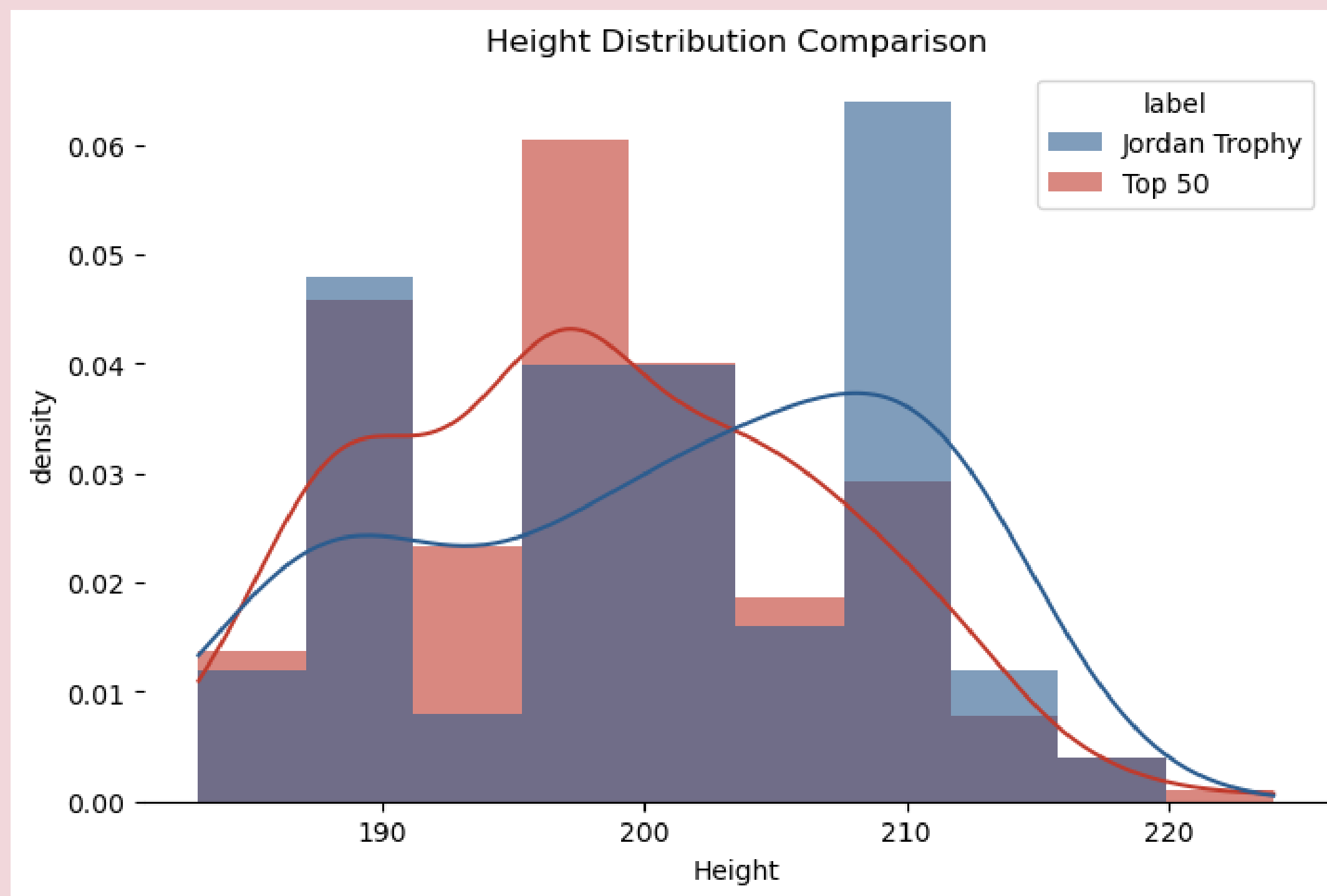
- balance ratio(ortg-drtg)



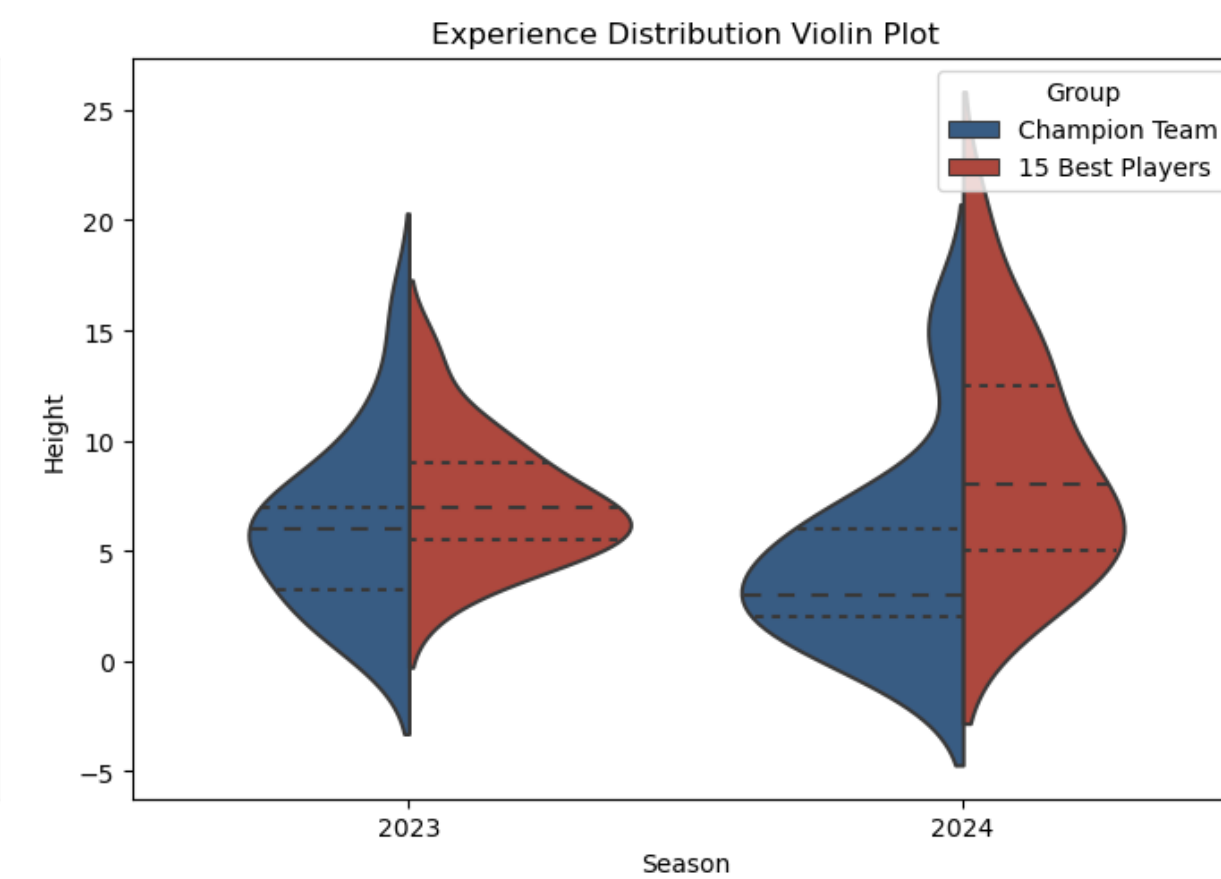
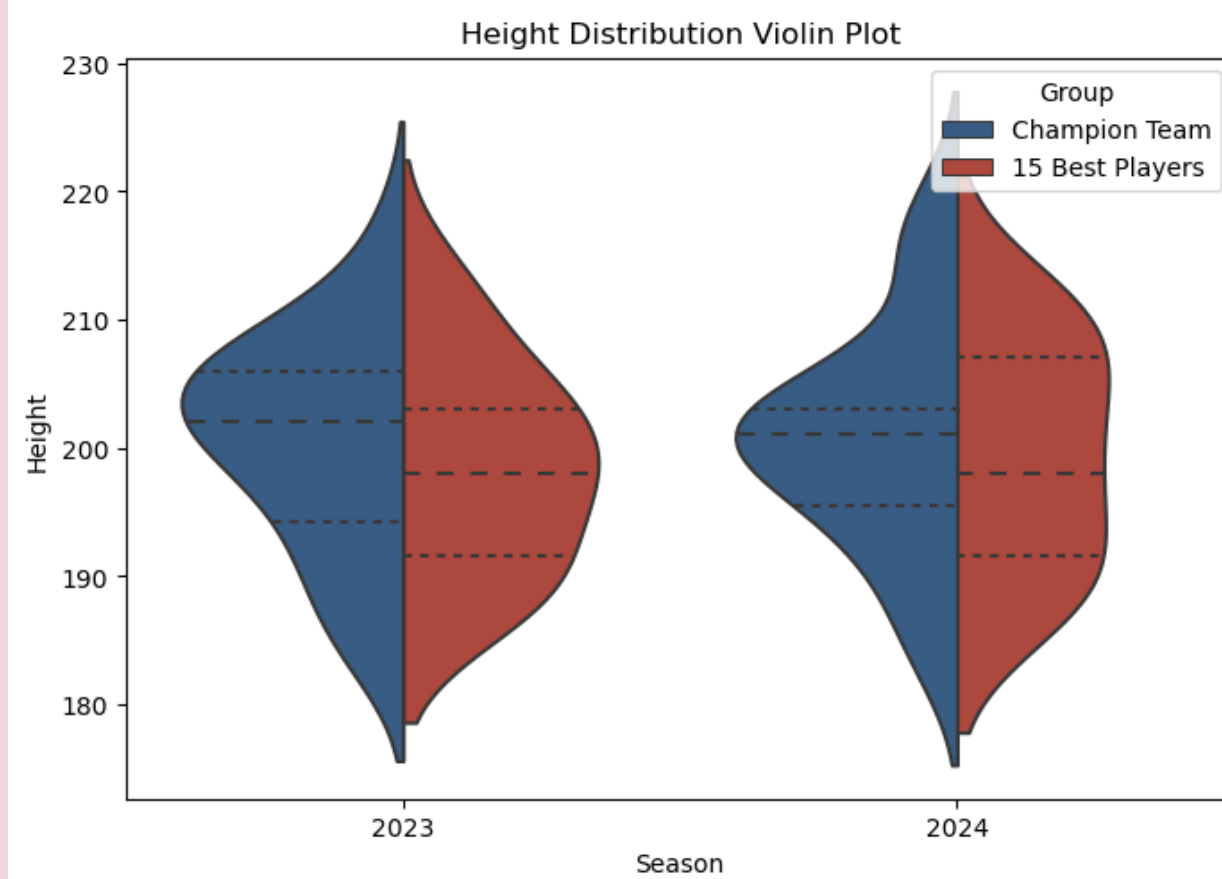
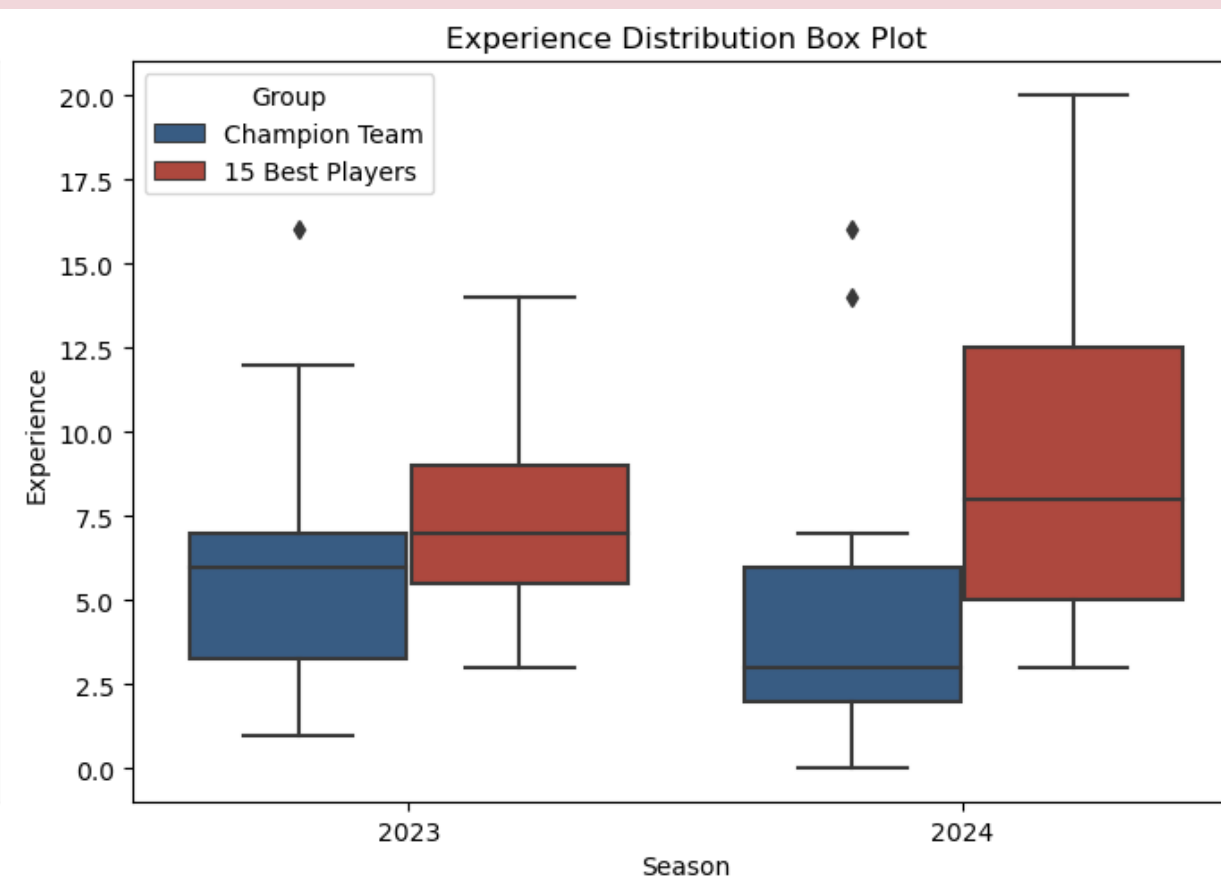
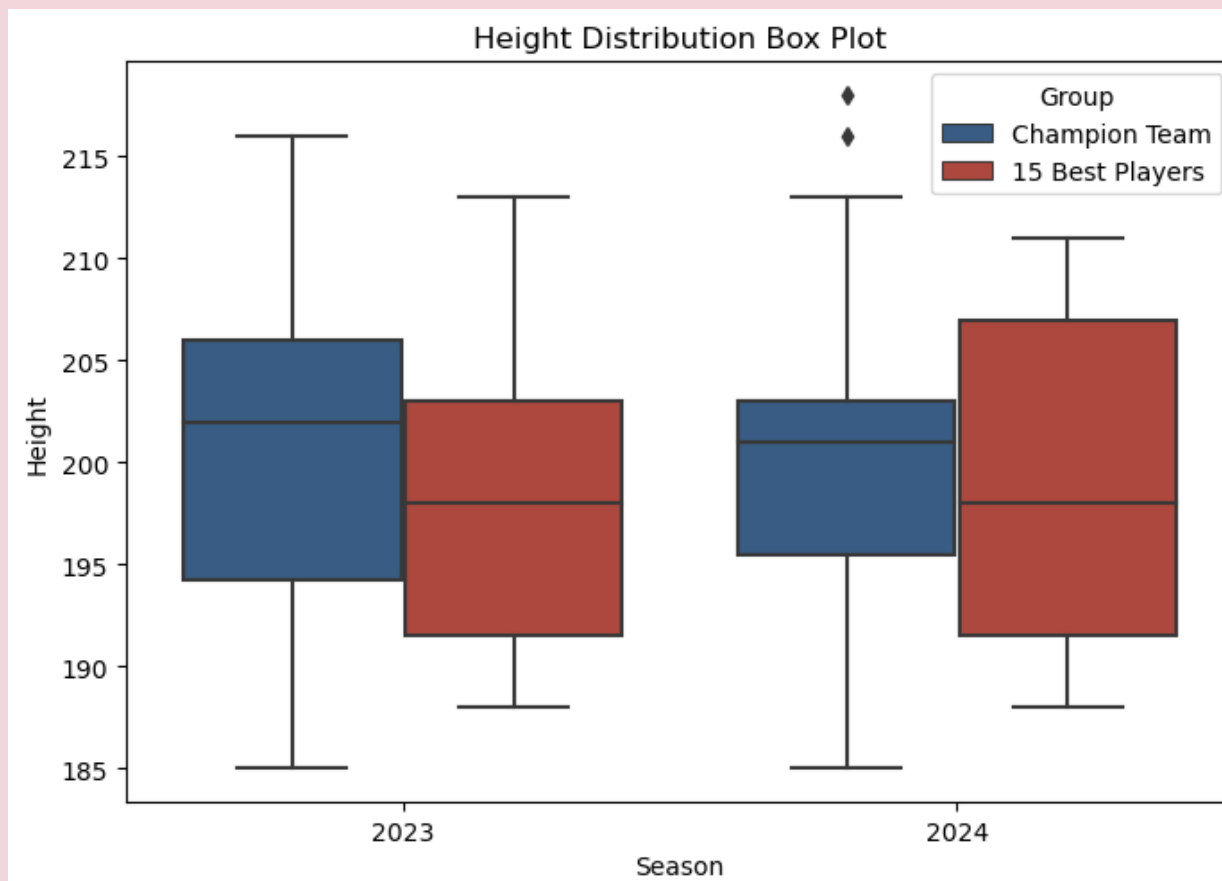


IT'S GAME TIME!

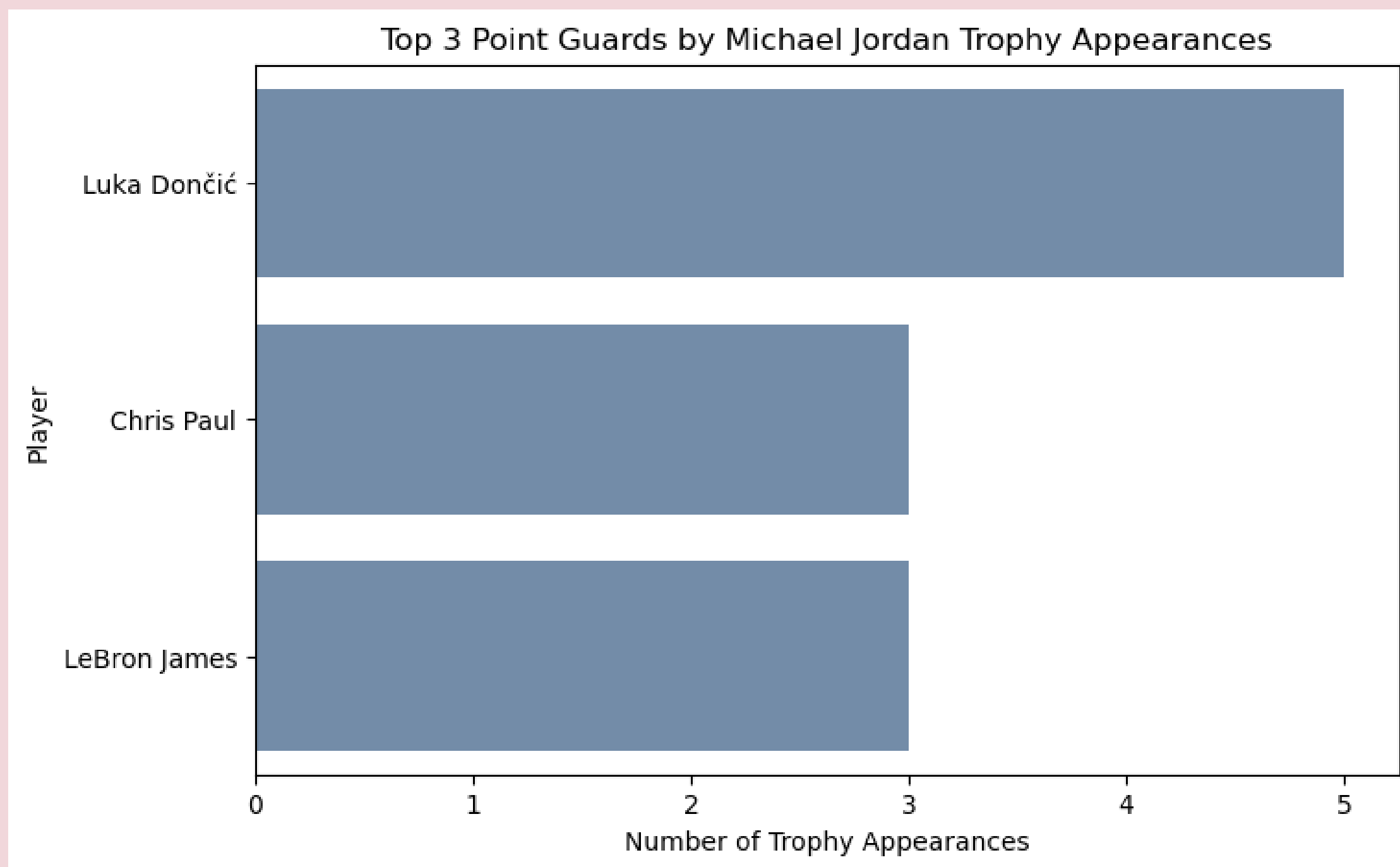
4th Quarter : Hypothesis Testing



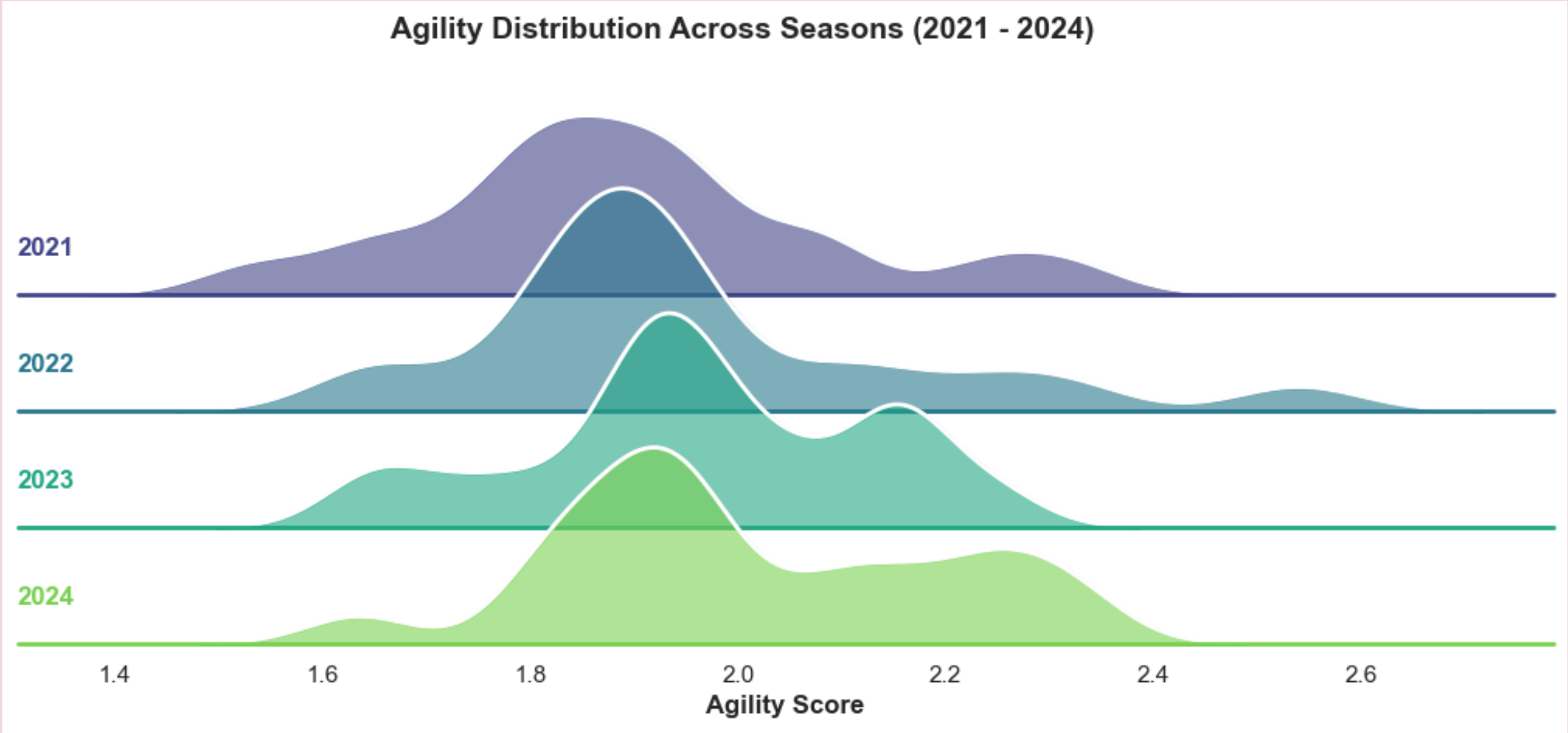
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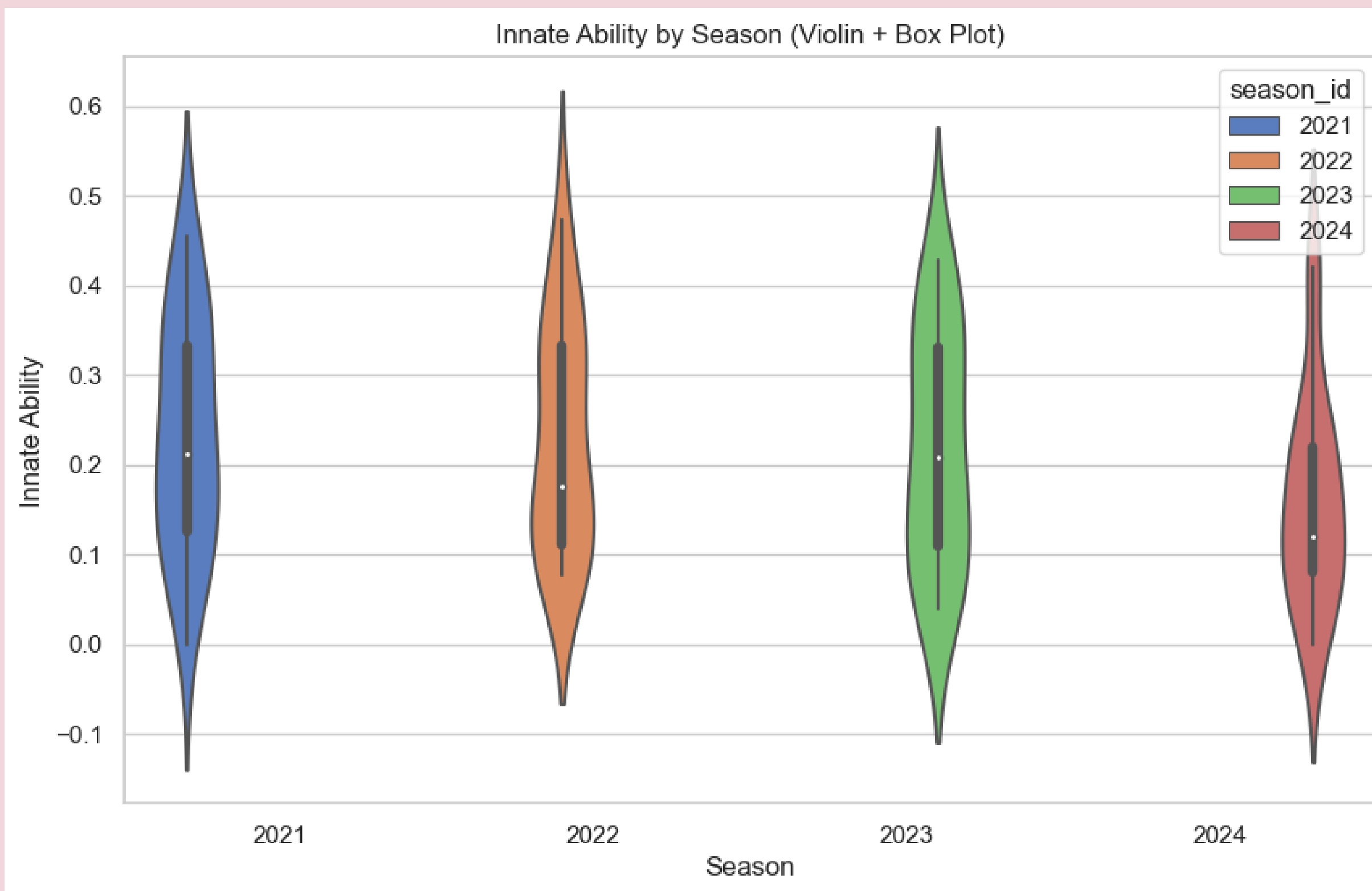
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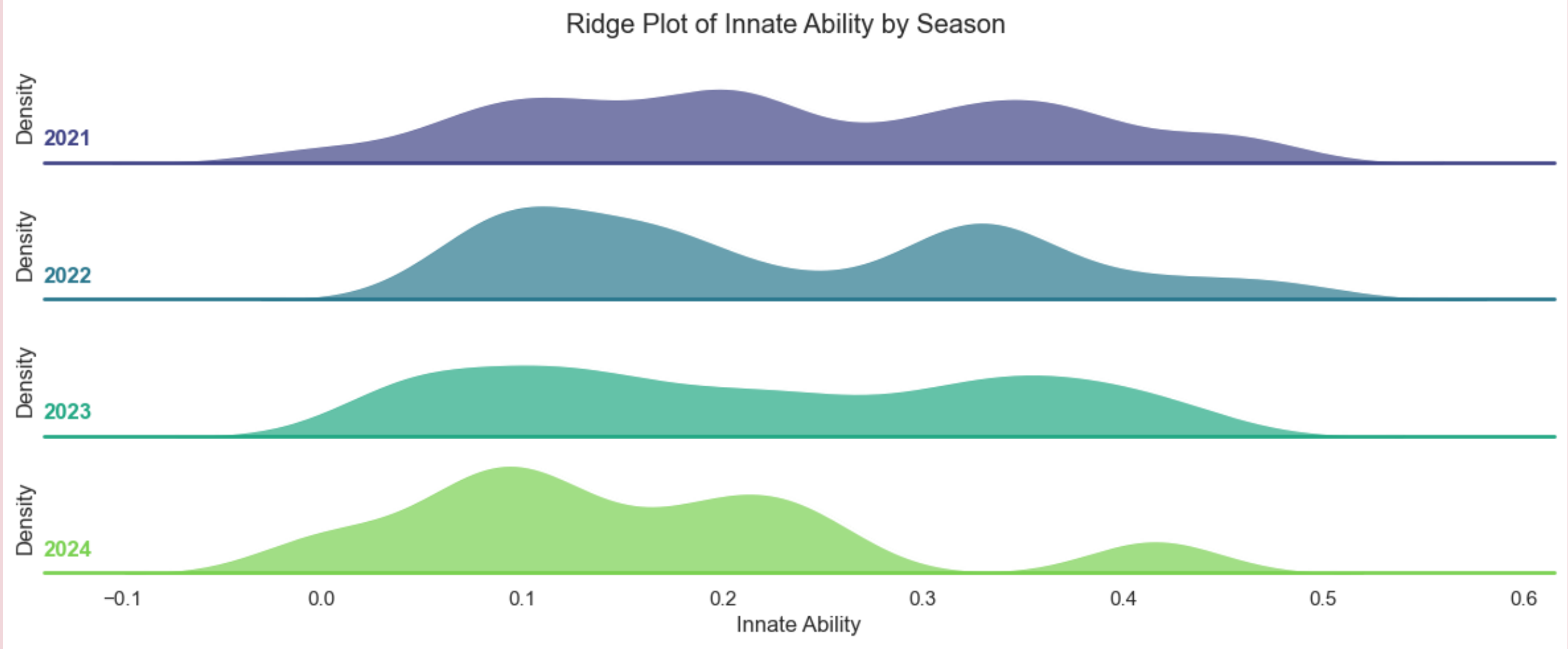
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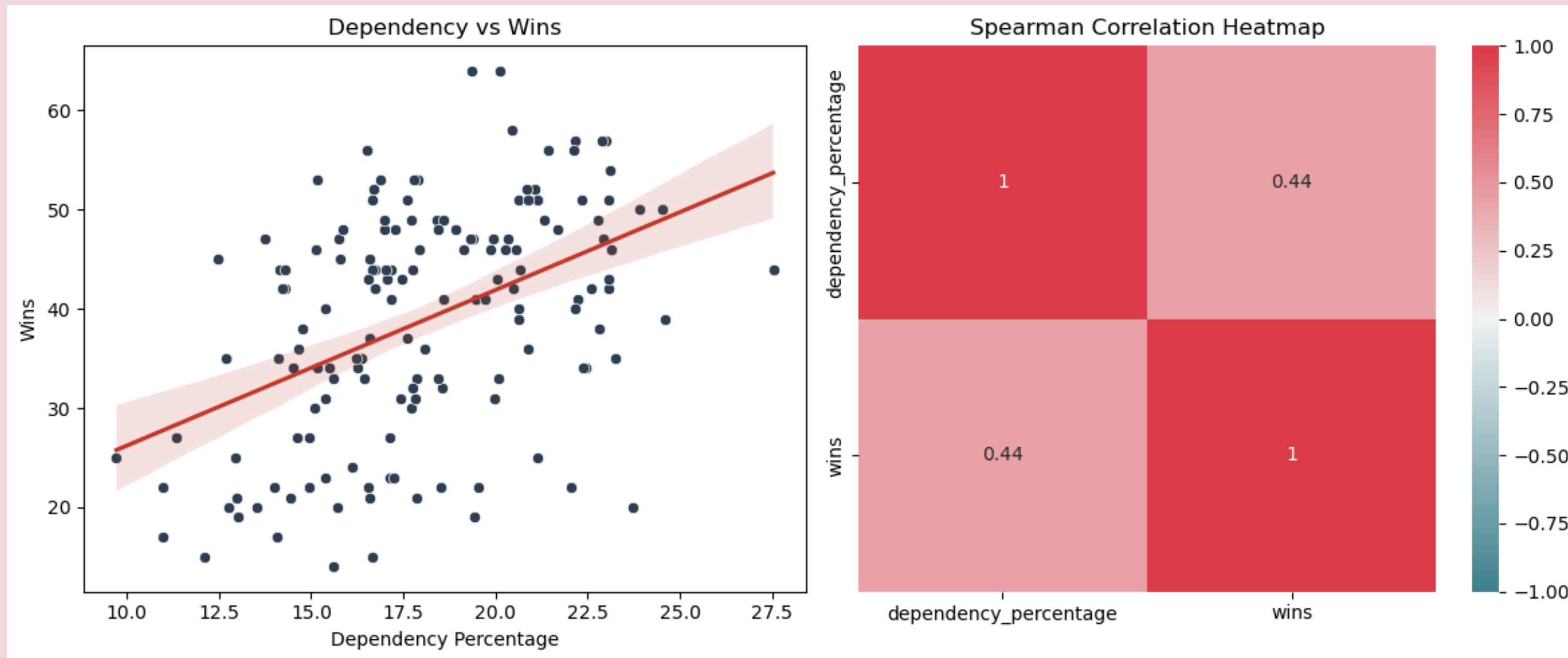
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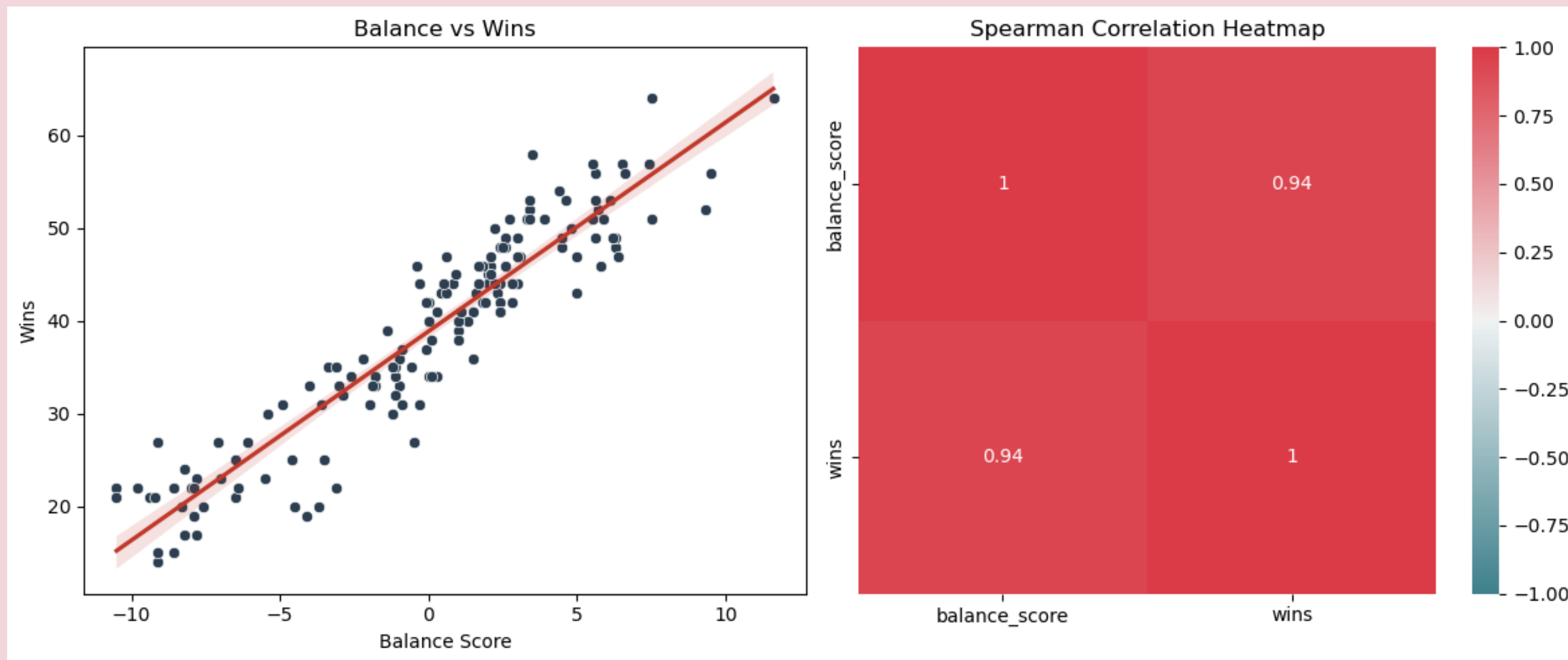
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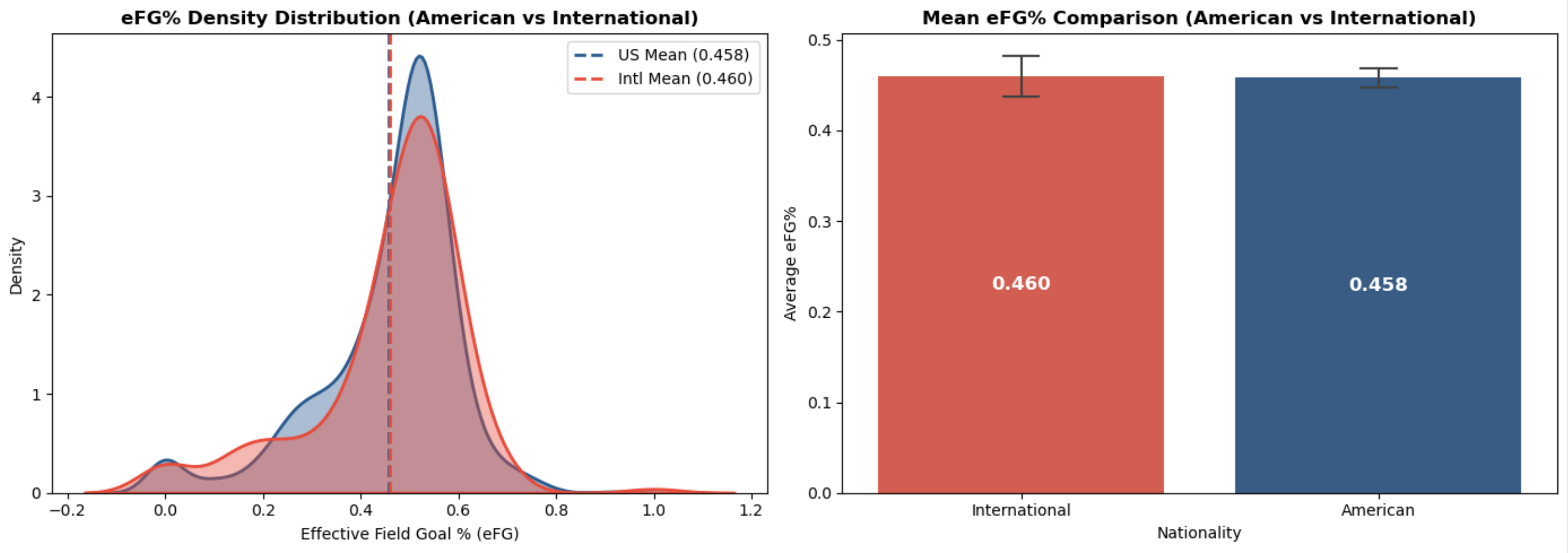
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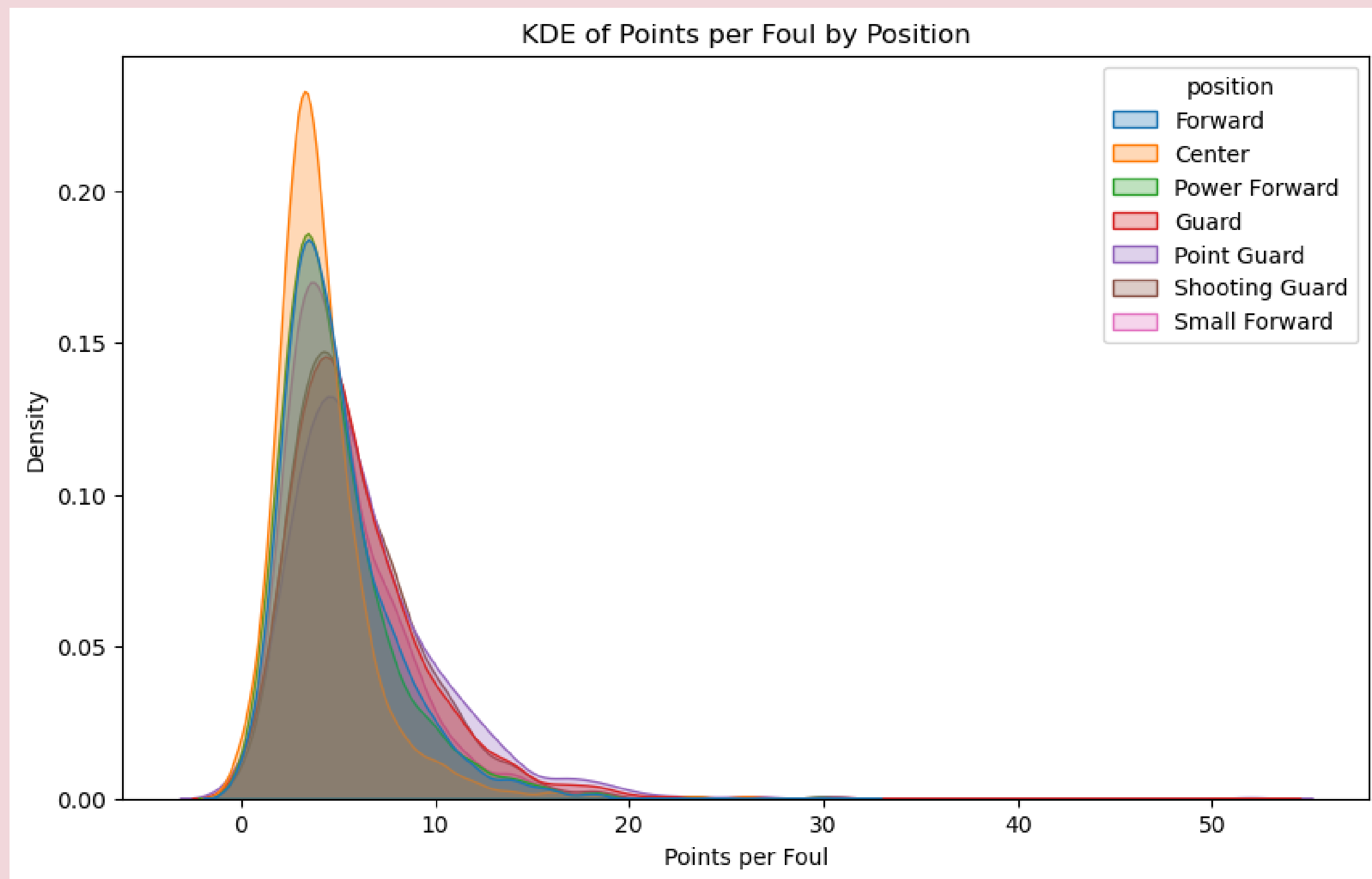
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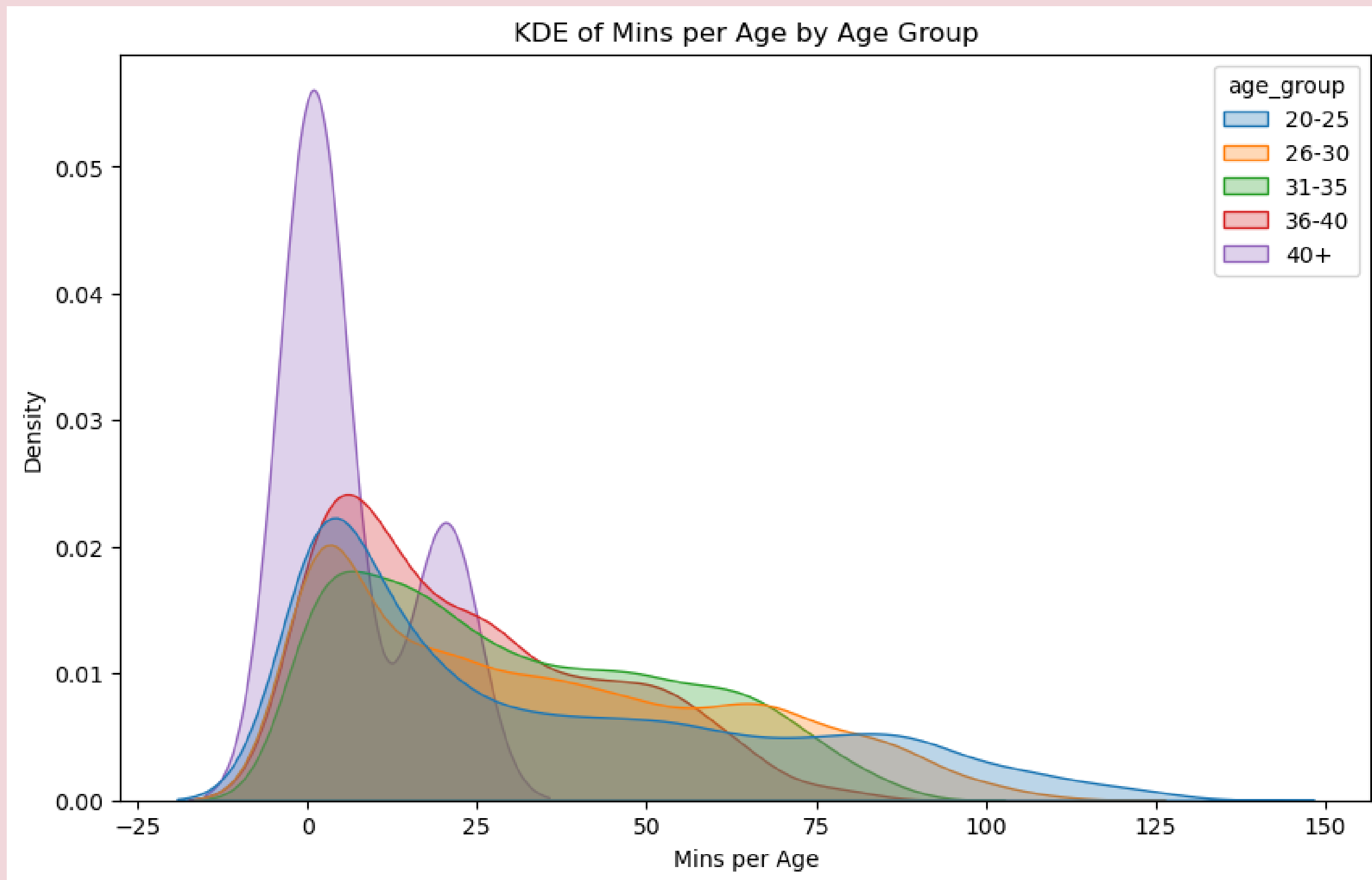
6-2



7-2



8-2



9-2